Introduction to Programming and Data Structures

Data Structures - Basics

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Basics

A data structure is a logical organization of data

- A data structure can accommodate one or more data elements (storage).
- The logical organization defines
  - how the elements will get added (insertion)?
  - how the elements will get removed (deletion)?
  - how the elements will be browsed (traversal)?

data structure + coding instructions = software/application

Types of Data Structures

- Linear
  - Stack
  - Queue
  - Linked List
- Nonlinear
  - Tree
  - Graph

array/list → index

hash table/map → key : value

- BIG O Notation (worst case, refers to very large value of n)
- keep fastest growing term , drop constants

example1:

```
for
    for
if list[k]==g
for
```

O(n^2)
O(1)
O(n)

== O(n^2)

example2:
binary search
kth iteration : n/2^k
0(log2(n))

Problems

ARRAYS (elements + index , continuous memory location)

can be multidimentional (nested lists)

1) static (fix length)
2) dynamic
$ allocate fix size
$ if required, allocate new memory
new size = 2*previous size
and copy the data from previous array

1. Can nonlinear data structures be represented pursuing a linear organization? (ans.) yes, example trees with no addtional branch, graphs with linear nodes only

lookup by index: O(1) ; list[k]
lookup by value: O(n) ; for list[i]==val
array traversal: O(n) ; for i in list
array insertion/deletion: O(n) (index changes/shifts) ;
list.insert/append/pop/delete

amortized (may require (_)